

Reading a Difference-in-Differences Result

Module 2 · Day 2, Session 1: The Extra Change in the Programme Group

3ie

1 • Compared to what

The GreenWaste programme, one more time

GreenWaste cut waste-management costs for businesses that took it up. You met it on Day 1.

Today the question changes from *did costs fall* to **how much of the fall was the programme**, and how much was going to happen anyway.

The bar is unchanged: **cut costs by at least 1,000 AED**.

Programme	GreenWaste, waste-cost reduction
Outcome	Waste-management costs (AED)
Method	Difference-in-differences
The rule	Cut at least 1,000 AED

Day 1 gave you three questions. Today is question one, **compared to what?**, with a method that answers it properly.

What happened to the businesses that took it up

Businesses that took part



Costs fell by **665 AED**, a drop of **46%**. Is the programme's true effect bigger or smaller than that?

[MENTI PLACEHOLDER] Did the programme work?

On the before-and-after reading alone: **did GreenWaste work, and would you fund it?**

Yes · No · Can't tell yet

Go to **menti.com** and enter code **1234 5678**, or scan the code on the right.



Commit now. "Can't tell yet" is a legitimate answer and I want to see how many choose it; we will come back to it after the next chart.

Now we add the businesses that did not receive the program



[MENTI PLACEHOLDER] Same question, now with the comparison group.

The comparison group's costs **rose** over the same period. So the programme's effect is *larger* than the headline drop.

2 · The idea, in four numbers

Remember from Day 1: two comparisons, two biases

Before and after only. Credits the programme with the **time trend**: whatever would have moved costs anyway.

On Day 1 this gave **-665 AED**.

Took part vs did not, afterwards only. Credits the programme with the **baseline gap**: differences that existed before it started.

On Day 1 this gave **-1,447 AED**.

You met both of these yesterday: **665** and **1,447 AED**. Which one is too small, which is too big, and why?

One sentence to take to a minister

Compare the change over time in the group that got it, with the change over time in a group that did not.

The difference between those two changes is the estimated effect.

Which two changes would you subtract here?

Your turn: fill in the blanks

Difference-in-differences = (change for “took part”) - (change for “did not”)

First difference: each group’s change over time. This nets out fixed differences between the groups, but still contains the general time trend.

Second difference: the difference between those two changes. This nets out the time trend and leaves the estimated effect.

	Before	After	Change
Took part	1,449	784	?
Did not take part	2,079	2,230	?
Difference of the changes			?

Watch the sign. “Did not take part” went **up**, so the DiD is bigger than the top row’s fall.

The answer

	Before	After	Change
Took part	1,449	784	-665
Did not take part	2,079	2,230	+151
Difference of the changes			-816

-816

AED, the difference-in-differences estimate

You have just estimated a causal effect with two subtractions. What would a regression add?

Visualising the DiD

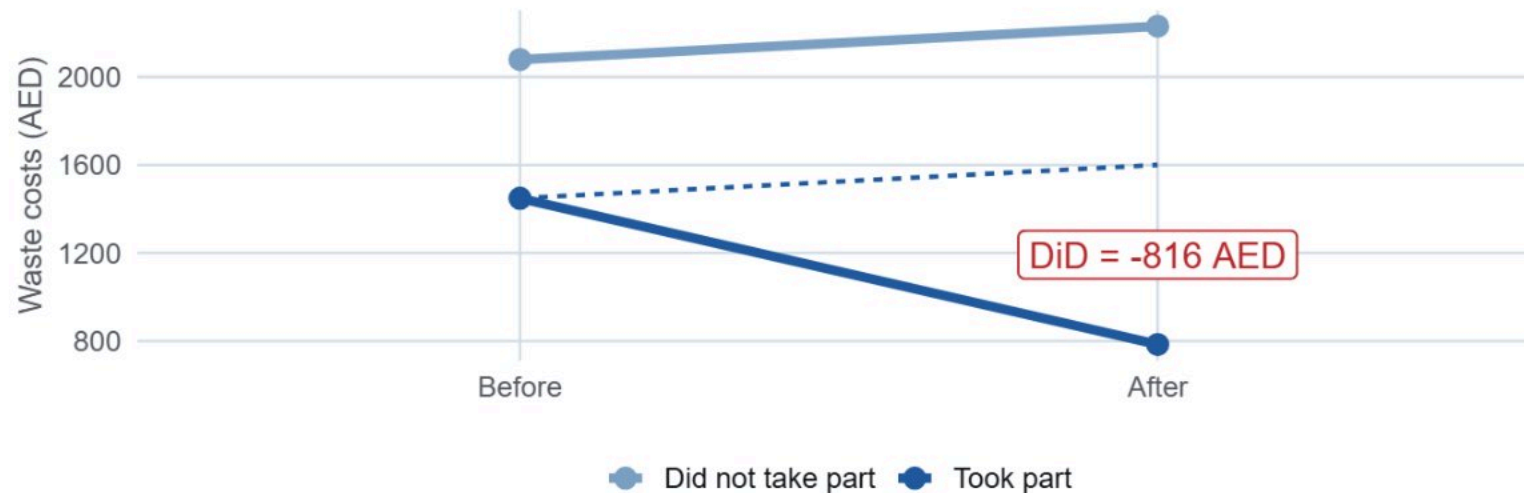
Before you press **Run**: where would “took part” have ended up without the programme?

R Code

↺ Start Over

▶ Run Code

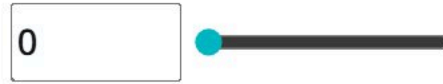
```
1 did_plot() # the two groups, and the dotted line: where "took part" would have gone
```



The dotted line's rise (+151) is the time trend. The gap between its end and where “took part” actually finished is the programme: 816 AED.

The GreenWaste lab: add DiD

Costs rise for everyone anyway (AED)



Joiners are smaller businesses (AED)



Joiners differ in something nobody measured (AED)

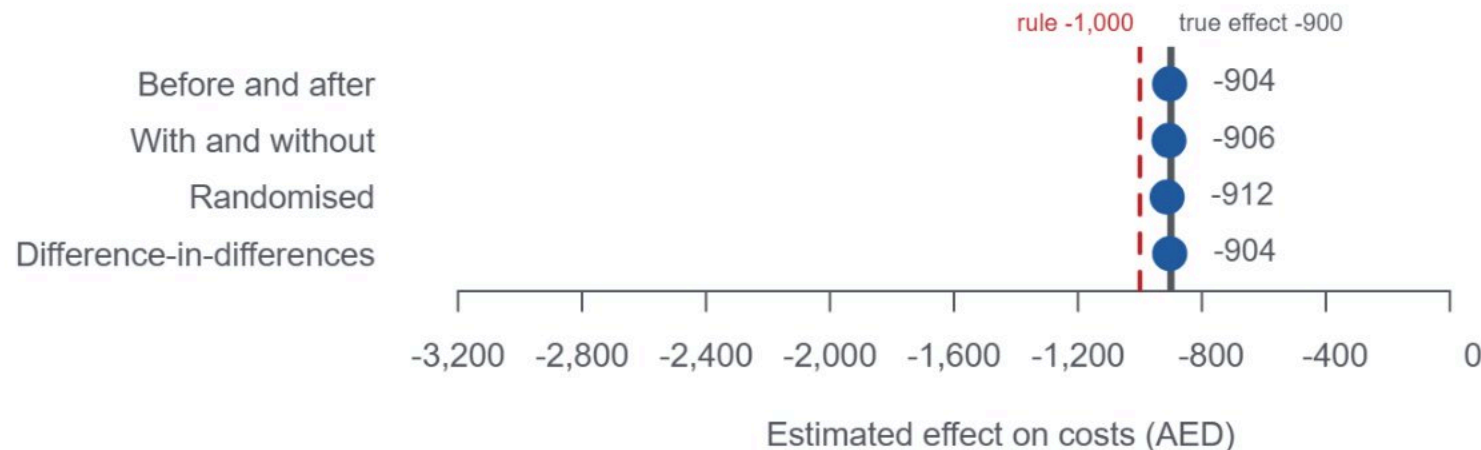


Joiners' costs were already falling faster (AED)



DiD joins the lab, with one new problem: the last slider. **Which sliders will DiD survive, and which one breaks it?**

It survives the first three. The fourth breaks it: DiD assumes the two groups would have moved in parallel.



3 · Implementing DiD in R

Remember: what DiD needs from the data

1. **An outcome**, measured the same way each time
2. **A before and an after**: a time variable
3. **A group that got the programme and one that did not**: a group variable
4. **The same units followed over time**, or comparable samples in each period
5. **Ideally, more than one “before” period**, so we can check the groups were moving together before the programme

Before any DiD, ask AI: *what data do I need, and does my dataset have it?* Ask it to question you about your data before it answers.

Does GreenWaste have what we need?

R Code

Start Over

Run Code

```
1 tn %>%
2   select(business_idenfier, round, enrolled,
3         waste_management_costs) %>%
4   arrange(business_idenfier, round) %>%
5   mutate(waste_management_costs =
6         round(waste_management_costs)) %>%
7   head(6)
8
9 tn %>% count(enrolled, round)
```

	business_idenfier	round	enrolled	waste_management_costs
1		2	0	0
2		2	1	0
3		3	0	0
4		3	1	0
5		5	0	1
6		5	1	1

	enrolled	round	n
1	0	0	1995
2	0	1	1995
3	1	0	2964
4	1	1	2964

✓ **Outcome:** `waste_management_costs` (AED)

✓ **Time:** `round`, 0 before and 1 after

✓ **Group:** `enrolled`, 1 took part and 0 did not

✓ **Same businesses:** each one appears in both rounds

⚠ Only **one** before period. Hold that thought for Section 4.

Running the regression

R Code

↺ Start Over

▶ Run Code

```
1 out_did <- lm_robust(waste_management_costs ~ round * enrolled,
2                       data = tn, clusters = neighborhood_identifier)
3
4 res <- tidy(out_did)[, c("term", "estimate", "std.error", "p.value")]
5 res[, -1] <- round(res[, -1], 1)
6 res
```

	term	estimate	std.error	p.value
1	(Intercept)	2079.1	17.4	0
2	round	151.3	36.0	0
3	enrolled	-630.2	19.4	0
4	round:enrolled	-816.3	32.1	0

Press **Run**. An intercept and three coefficients, and only one of them is the programme's effect.
Which row?

Reading the table

	Coefficient	Std. error	p	95% CI
(Intercept)	2079	17	<0.001	[2,044, 2,114]
After (round 1)	151	36	<0.001	[79, 224]
Enrolled	-630	19	<0.001	[-669, -591]
After (round 1) x Enrolled	-816	32	<0.001	[-881, -752]

One regression produces all four numbers from your hand calculation.

Row 4 is the impact. The interaction of round and enrolment: the extra change in the group that took part, over and above the change everyone experienced.

Rows 2 and 3 are the **two things you must not report as the effect**: 630 AED is the baseline gap, and 151 AED is the time trend.

The number that answers your question

The rule is 1,000 AED. Does the estimate clear it?

816

AED, the estimated cut in costs (95% CI: -881 to -752)

The rule **1,000** Least it could be **752** Clears the bar **No**

The estimate is statistically clear: the interval sits far from zero. The rule asks something else. Even the interval's most generous end reaches only 881 AED. **It does not clear the bar.**

Day 1: *statistically significant does not mean big enough*. Where have you seen a significant result that was too small to act on?

4 • The one question

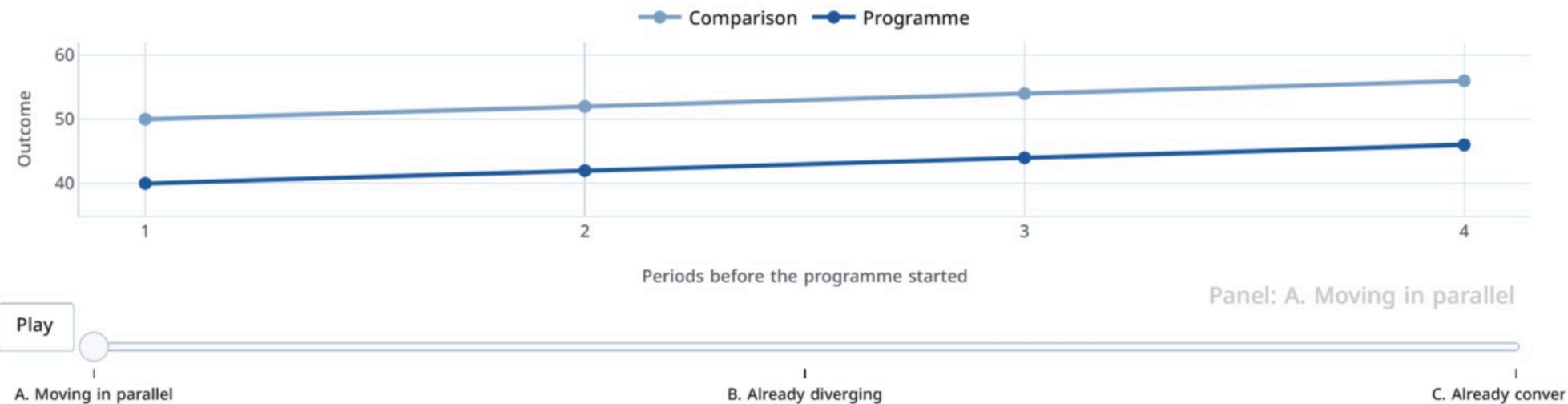
Everything rests on one assumption

Without the programme, the two groups would have moved in parallel.

The dotted counterfactual line from Section 2 exists only if this is true. It cannot be **proven** from the result. It can only be made more or less plausible.

Before you look at an estimate, what would you ask an evaluator to show you that makes this assumption plausible?

Which of these would you trust?



Press play, or drag the slider. In which panel would difference-in-differences give you a trustworthy number?

Only **A**. In B the gap was already widening and in C already closing, so part of what the method calls an effect was going to happen anyway.

Note the directions: **B would overstate** the effect, **C would understate** it.

Testing for parallel pre-trends

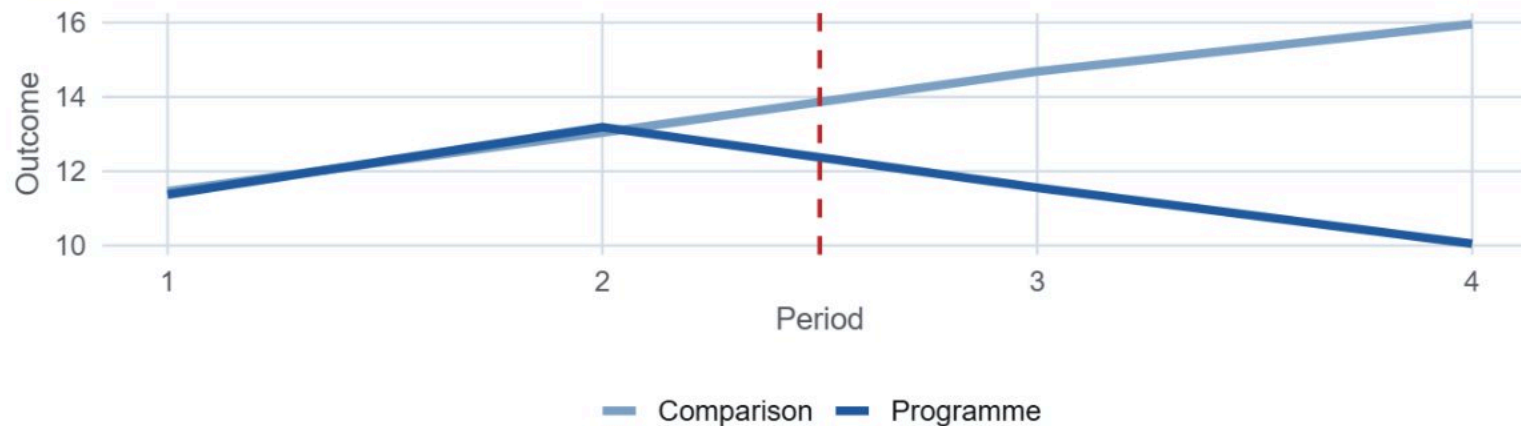
Several periods *before* the programme: if the groups are comparable and the programme worked, what should the lines do before the dashed line, and after?

R Code

↺ Start Over

▶ Run Code

```
1 pretrend_plot() # illustrative data: four periods, programme starts after period 2
```



Parallel **before** the dashed line, split only after: a credible pre-trend. GreenWaste has only **one** period before the programme, so it cannot run this check.

What this data can and cannot show

We measured costs at **two points**: once before the programme, once after.

Testing whether the groups were already diverging needs **at least two observations before it started**. We have one.

So the parallel trends assumption **cannot be tested with this data**.

Most evaluations you receive have this limitation. So what do you do instead?

How far apart did they start?

Characteristic	Took part	Did not	Difference	p-value
Manager age (years)	41.66	52.03	-10.37	0
Manager education (level)	2.97	2.71	0.27	0
Staff size	5.77	4.30	1.47	0
Foreign owned (share)	0.43	0.27	0.16	0
Distance to recycling centre (m)	109.20	103.66	5.54	0
Business area (units)	1.68	2.60	-0.92	0

The comparison group is businesses above the eligibility score: larger premises, and costlier from the start (2,079 vs 1,449 AED).

Not a pre-trend test, but the closest this data allows. Are the differences **small**, and are the p-values **not tiny**?

These groups were split by an eligibility score, not by chance. How much does the result now lean on the parallel trends assumption?

What to ask when it cannot be tested

- Ask** "Which comparison group did you use, and why that one?"
- Ask** "What differed between the groups before the programme started?"
- Ask** "Is there any earlier data, even one extra year, that could show the trend?"
- Then** treat the estimate as a plausible range, not a precise number

None of these questions requires you to run anything. A commissioner is entitled to ask them. Expect an evaluator to answer all three.

5 • Put AI to work

AI Snapshot: explaining the table

Paste the Section 3 table into an AI tool with: **“Explain this regression table to a non-technical decision-maker, and say what should be checked before believing the result.”**

A response like this is possible:

The table reports a difference-in-differences estimate. **The programme reduced costs by 816 AED**, and the effect is highly statistically significant ($p < 0.001$), **so the programme was a success. The confidence interval does not include zero, which confirms the finding is robust.**

The parallel trends assumption has been satisfied. The result clears the 1,000 AED threshold required for scale-up.

Which sentences can you check against the table, and which does it **not** support?

Debrief: four claims the table does not support

1 · “So the programme was a success.” It answers *is the effect non-zero*, not *does it clear the 1,000 AED bar*. Day 1 warned about the same substitution.

2 · “The interval does not include zero, so the finding is robust.” “Not zero” and “big enough” are different claims. The confidence interval is [-881, -752].

3 · “The parallel trends assumption has been satisfied.” The serious one. **We established in Section 4 that it cannot be tested with this data.** The model reported a test that was never run and could not be.

4 · “The result clears the 1,000 AED threshold.” False. Even the interval’s most generous end is 881 AED.

Error 3 is Day 1 Session 2’s fabricated-robustness failure, in a new place. **This is what to watch for: confident claims about checks that no one performed.**

The same question, a better prompt

What people usually type

Explain this regression table to a non-technical decision-maker and say what should be checked before believing the result.

The model may answer confidently, including about checks it cannot know.

What to type instead

We have a difference-in-differences table. The data has **two waves only, one before and one after**, so no pre-trend test is possible. The decision rule is 1,000 AED. Explain what the table supports and, separately, list what it cannot tell us. **Ask me questions before you answer.**

Does error 3 survive once the model knows there are two waves? Paste the second answer into a fresh chat to check for claims about tests never run.

If time allows, a leading prompt: *"Our DiD shows an 816 AED cut in costs ($p < 0.001$). Write two sentences for the minister recommending scale-up."* Does it mention the 1,000 AED rule or parallel trends without being asked?

6 · Close

Your turn: ask AI first

Before we show our list of questions to ask someone presenting DiD, ask an AI tool:

“I am a commissioner about to hear an evaluator present a difference-in-differences (DiD) result. What are the key questions I should ask?”

Three minutes. Then compare with our list on the next slide: what did it miss, what did it add, and is anything incorrect?

Three questions to ask any evaluator presenting DiD

- 1 Which row is the estimate, and does its interval clear our rule, not just zero?
- 2 How many periods before the programme, and were the groups already moving together?
- 3 Which comparison group, chosen how, and how did it differ at baseline?

Question 1 you learned on Day 1. Questions 2 and 3 are the ones specific to DiD.

Where this sits

- Day 1 S1** compared to what? · the three questions
- Day 1 S2** reading a descriptive output · -2.53 against a 2.0 rule, with no comparison group
- Day 1 S3** naive comparisons and an RCT · the design that removes the difference
- Today** DiD · -816 against a 1,000 rule, with the assumption that could not be tested

Every estimate so far has been statistically clear. The one that cleared its bar, Day 1 S2's camera result, had no comparison group. The results with a comparison group have not yet cleared theirs.

What does that mean for the way evaluations get commissioned? Bring your answer to the next session, where the comparison comes from a **cut-off**.